2/19/2019

Ex: Ideal Ramjet Engine : Consider a ramjet powered interceptor missile flying @ $M = 3.5$, 10km altitude. It develops thrust of 1 kN & consumes 60 grams/sec of fuel (HC) with $H_{vf} = 43\ kJ/g$. This $\dot{m}_f$ is sufficient to increase stag. temp by 100%. Determine f, η_o, η_{th}, η_p.

$$f : \frac{\dot{m}_f}{\dot{m}_a}$$

$(\dot{m}_f = 0.06\ kg/s)$; $\dot{m}_a = ?$

$h = 10\ km$; $T_o = 223K$.

$$\boxed{C_p(T_{t2} - T_{t1}) = \underline{q} = \underline{\underline{\dot{m}_f H_{vf}}}}$$

{Assume $\dot{m}_{in} = \dot{m}_{out}$
not true b/c $\dot{m}_{in} = \dot{m}_a$
$\dot{m}_{out} = \dot{m}_a + \dot{m}_f$}

↳ Cannot use this.

$$T_{t2} = 2 \times T_{t1} = 2 \times T_o \left(1 + \frac{\gamma - 1}{2} M_o^2\right)$$

$M_o \to 3.5$

$T_{t1} = 769K.$

$T_{t2} = 1538K.$

© North Carolina State University, All Rights Reserved

1st Law for combustor

$$\dot{Q} - \cancel{\dot{W}}^{\,0} = \dot{m}\left(\Delta h + \cancel{\Delta KE} + \cancel{\Delta PE}^{\,0}\right)$$

$$\dot{m}_f H_{vf} = \dot{m}_{out} h_{t,out} - \dot{m}_{in} h_{t,in}$$

$$= (\dot{m}_f + \dot{m}_a) C_p T_{t2} - \dot{m}_a C_p T_{t1}$$

$$\left\{ \begin{array}{l} h_t = h + KE \\ \Delta h_t = \Delta h + \Delta KE \end{array} \right\}$$

$$\dot{m}_f (H_{vf} - C_p T_{t2}) = \dot{m}_a C_p (T_{t2} - T_{t1})$$

$$\dot{m}_a = \frac{\dot{m}_f (H_{vf} - C_p T_{t2})}{C_p (T_{t2} - T_{t1})} =$$

$$= \frac{0.06\,(43\times10^6 - 10^3\times1539)}{10^3(1539 - 769)} =$$

$$\boxed{\dot{m}_a = 3.2 \text{ kg/s}}$$

$$f = \frac{\dot{m}_f}{\dot{m}_a} = \frac{0.06}{3.2} = \underline{\underline{0.019}}$$

$$(f_{st,HC} = 0.06)$$

$$\because \dot{m}_a = 3.2 \text{ kg/s} \Rightarrow \frac{\tau}{\dot{m}_a} = \frac{10^3}{3.2} = 3125 =$$

$$\frac{\tau}{\dot{m}_a} = (1+f)u_e - u_0$$

$$\Rightarrow u_e = \frac{\frac{\tau}{\dot{m}_a} + u_0}{(1+f)} = 1050 \text{ m/s.}$$

$$u_0 = 3.5 \times \sqrt{\gamma R T_0} = 223$$

© North Carolina State University, All Rights Reserved

$$u_e = \frac{312 + 1050}{1.019} = 1336 \text{ m/s.}$$

$$u_0 = 1050 \text{ m/s} \qquad \left(u_0 = M_0 \sqrt{\gamma R T_0} \right)$$

$$\eta_{th} = \frac{(1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2}}{f(H_{vf})} = 0.44$$

$$\eta_p = \frac{\overset{\to 10^3}{\tau} \overset{\to 1050}{u_0}}{\underset{\hookrightarrow 3.2}{\dot{m}_a} \left((1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2} \right)} = 0.91.$$

$$\eta_0 = \eta_{th} \cdot \eta_p = \underline{\underline{0.40}}$$

© North Carolina State University, All Rights Reserved

Ex 2: Hydrogen fueled Scramjet flying @ $M = 7$, $h = 30$ km. Fuel burned @ overall equiv. ratio of 0.7 with propulsive efficiency of 90%. Determine the specific thrust, TSFC, η_{th}, η_o

$M = 7.0$, $h = 30$ km $\quad \frac{T}{T_{SL}} = 0.7 \Rightarrow \underline{\underline{T = 220K}}$

$$u_0 = \sqrt{\gamma R T_0}\, M = 7 \times \sqrt{\gamma R T_0}$$

$$u_0 = 2100 \text{ m/s.}$$

Specific thrust: $\frac{\tau}{\dot{m}_a} = (1+f)u_e - u_0$

Unknowns: f, u_e.

$$\phi = \frac{f}{f_{st}}$$

$$f_{st} \text{ for } C_xH_y = \frac{36x + 3y}{103(4x+y)}$$

{ here for H_2 $x=0; y=2$ }

$$f_{st} = \frac{6}{206} = 0.03$$

$$\Rightarrow f = f_{st} \times \phi = \underline{\underline{0.02}}$$

© North Carolina State University, All Rights Reserved

$$\eta_p = \frac{u_0\left\{(1+f)u_e - u_0\right\}}{(1+f)\frac{u_e^2}{2} - \frac{u_0^2}{2}}$$

$$f \approx 0 \Rightarrow 1+f \approx 1$$

$$\eta_p \approx \frac{2}{\frac{u_e}{u_p}+1} = 0.9$$

$$\Rightarrow u_e = u_0\left(\frac{2}{\underbrace{\eta_p}_{0.9}} - 1\right)$$

$$u_e \approx 2540 \text{ m/s.}$$

$$\frac{\tau}{\dot{m}_a} = (1+f)u_e - u_0 = (1+0.02)\times 2540 - 2100 = 490 \text{ Ns/kg.}$$

$$TSFC = \frac{\dot{m}_f}{\tau} = \frac{f\dot{m}_a}{\tau} = \frac{0.02}{490} = 40e^{-6} \text{ kg/N.s.}$$

$$I_{sp} = \frac{\tau}{\dot{m}_f g} = \frac{\tau}{f\dot{m}_a g} = \frac{1}{TSFC}\cdot\frac{1}{\underbrace{g}_{10\,m/s^2}} = 2500 \text{ s.}$$

$$\eta_o = \frac{(1+f)u_e u_0 - u_0^2}{f H_{vf}} = \underline{\underline{0.44}} \quad \left(H_{vf} \text{ of } H_2 = 120\times10^3 \text{ J/kg}\right)$$

$$\eta_{th} = \frac{\eta_o}{\eta_p} = \underline{\underline{0.5}}$$

© North Carolina State University, All Rights Reserved

In Prob 22 (HW), k shd be 0.25 NOT 0.5

Range formula

Masses

Define m_i = initial mass of vehicle.

$$= m_p + m_e + m_f$$

m_p : payload mass

m_e : empty mass (structure)

m_f : fuel mass.

Thrust

Uninstalled thrust $= \tau$.

When we install in aircraft net thrust will be less due to installation loss. Lump all together into a new term called (D_e) external engine drag.

Define Installed thrust $\hat{\tau} = \tau - D_e$

© North Carolina State University, All Rights Reserved

D_e will be a funct^n of V & thrust reqd will also be $f(V)$, so D_e will be a funct^n of τ

Assume a linear relation such that

$$D_e = \phi_e \tau$$

Thus

$$\hat{\tau} = \tau(1-\phi_e)$$

<u>Cruise conditions</u>

When an aircraft is @ Cruise, then drag is exactly offset by thrust, such that the aircraft is not accelerating

$$\hat{\tau} = D = \tau(1-\phi_e) \quad @ \text{ cruise}$$

Recall definition of overall efficiency

$$\eta_o = \frac{\tau V_o}{\dot{m}_f H_{vf}}$$

Substitute & rearrange

$$\hat{\tau} = \frac{1}{V_o}\eta_o \dot{m}_f H_{vf}(1-\phi_e)$$

& $\dot{m}_f = \frac{dm_f}{dt} = -\frac{dm}{dt}$ (Rate at which vehicle ~~loses~~ loses its mass)

© North Carolina State University, All Rights Reserved

So $\hat{\tau} = \dfrac{\eta_o H_{vf} (1-\phi_e)}{V_o} \dfrac{dm_f}{dt} = D$ @ cruise.

Also @ cruise, lift is exactly = weight

Neglecting centrifugal forces

$$m g_o = L \quad @ \text{ cruise}$$

Introduce drag & lift Coeff.

$$C_L = \frac{L}{\frac{1}{2}\rho_o V_o^2 S} \; ; \quad C_D = \frac{D}{\frac{1}{2}\rho_o V_o^2 S}$$

$$= \frac{L}{q_o S} \qquad\qquad = \frac{D}{q_o S}$$

S: some characteristic area of aircraft.

Let $\mathbb{R}$ be the range (distance)

$$V_o = \frac{d\mathbb{R}}{dt}$$

$$\dot{m}_f = -\frac{dm}{dt}$$

For cruise, substitute for m, cancel out dt term,

$$\boxed{\frac{dm}{m} = -B\, d\mathbb{R}} \quad \text{Breguet Range eq}^{n}$$

$$B = \frac{g_o}{\eta_o H_{vf} (1-\phi_e) \frac{C_L}{C_D}}$$

© North Carolina State University, All Rights Reserved

During cruise

$B \approx$ const $\Rightarrow$ we can easily integrate the above eq^n

$$\frac{m_{final}}{m_{init}} = \exp(-BR)$$

In terms of fuel mass m_f

$$m_f = \cancel{m_{fuel} - m_{final}}\ m_{init} - m_{final}$$

$$\Rightarrow \frac{m_f}{m_i} = \frac{m_{init} - m_{final}}{m_{init}} = 1 - \frac{m_{final}}{m_{init}} = 1 - \exp(-BR)$$

Define fuel mass fraction $\Pi_f = \frac{m_f}{m_i}$

$$\Pi_f = 1 - \exp(-BR)$$

$$\Pi R = -\frac{1}{B} \ln(1 - \Pi_f)$$

To get long range, we want ~~large~~ ~~small~~ huge Π_f, & a small $B \rightarrow$ large η_o, high H_{vf}, high C_L/C_D, low ϕ_e

© North Carolina State University, All Rights Reserved

Ex: Picking up the scramjet examples earlier, let $(1-\phi_e)\frac{C_L}{C_D} = 4.0$

$m_i = 10000\,kg$.

How much fuel is needed for a range of 6000 km?

$$\Pi_f = 1 - \exp(-BR)$$

$$B = \frac{\overset{\sim 10\,m/s^2}{g_0}}{\underbrace{\eta_0}_{0.5}\underbrace{H_{vf}}_{120\times10^6\,J/kg}\underbrace{(1-\phi_e)\frac{C_L}{C_D}}_{4}} = 4.17\times10^{-8}\,m^{-1}$$

$$\Pi_f = 1 - \exp(-\underbrace{B}_{4.7\times10^{-8}}\underbrace{R}_{6\times10^6})$$

$$\Pi_f = 0.22 \quad \& \quad \Pi_f = \frac{m_f}{m_i}$$

$$\Rightarrow \boxed{m_f = 2200\,kg}$$

© North Carolina State University, All Rights Reserved